

Lost in Retraining: Closed-Loop Learning and Model Collapse in Exponential Families

Fariba Jangjoo¹, Giovanni di Sarra¹, Matteo Marsili², and Yasser Roudi³

¹*Kavli Institute for Systems Neuroscience and Centre for Algorithms in the Cortex, Faculty of Medicine and Health Sciences, Norwegian University of Science and Technology, Trondheim, Norway*

²*ICTP- The Abdus Salam International Centre for Theoretical Physics, Trieste, Italy*

³*Department of Mathematics, King's College London, London, United Kingdom*



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Closed-loop learning is the process of repeatedly estimating a model from data generated from the model itself. It is receiving great attention due to the possibility that large neural network models may, in the future, be primarily trained with data generated by artificial neural networks themselves. We study this process for models that belong to exponential families, deriving equations of motion that govern the dynamics of the parameters. We show that maximum likelihood estimation of the parameters endows sufficient statistics with the martingale property and that as a result the process converges to absorbing states that amplify initial biases present in the data. However, we show that, for exponential families, this outcome may be prevented if, at each closed-loop retraining iteration, the data contains at least one data point generated from a ground truth model, by relying on maximum *a posteriori* estimation or by introducing regularization.

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Introduction—This Letter focuses on the behavior of statistical models and learning machines subject to *closed-loop* learning: the process in which models are repeatedly trained on data mainly generated by models themselves. When does this process converge and if so to what? These questions have raised considerable recent interest in the context of large language models (LLM) and other state-of-the-art neural networks [1–8]. The success of LLMs relies on the availability of huge amounts of high-quality human-generated data. Yet the supply of such data has been argued to be running out [9]. At the same time, more and more AI-generated data is populating the Internet, thereby polluting the datasets with which models will be trained. It has been argued that this process of “data-autophagy” is pathologic with a severity that depends on the fraction of synthetic data used in training [1,2,6]. A possible outcome of this is *model collapse* [4]—a phenomenon observed also in generative adversarial networks [10], diffusion models [11], variational autoencoders [4], and LLM [5]—whereby the model generates a limited variety of outputs. Yet Marchi *et al.* [3] find that closed-loop learning can also lead to the opposite outcome, where models generate random outputs. This shows that empirical evidence on the ultimate fate of closed-loop learning is unclear, not least because of the

large variety and complexity of the architectures employed. For this reason, here, we focus on analyzing closed-loop learning in simplified but powerful settings, namely the exponential family (EF) [12,13].

For a random variable s , we thus consider a distribution

$$f(s|\boldsymbol{\theta}) = \frac{1}{Z(\boldsymbol{\theta})} e^{\sum_a \theta_a \phi_a(s)}, \quad Z(\boldsymbol{\theta}) = \sum_s e^{\sum_a \theta_a \phi_a(s)}, \quad (1)$$

where $\boldsymbol{\theta} \in \mathbb{R}^D$ is a vector of D parameters with components θ_a . Each $\phi_a(s)$, $a = 1, \dots, D$, is a bounded function of s , and Z is the partition function. We aim to understand how the parameters evolve if they are fitted to data generated by the model with the previous set of parameters, including potentially data generated by an external source.

We first study the dynamics of model parameters when they are repeatedly estimated using maximum likelihood (ML) on data exclusively drawn from the model itself. For ML, the relevant variables are the sufficient statistics of the model [14], which, as we show, acquire a martingale property in the retraining dynamics. This property in turn will be shown to lead to model collapse. We then discuss how this collapse may be avoided. To do this, we consider the more general case in which parameter learning may go beyond ML and may involve a prior distribution/regularization, as in maximum *a posteriori* (MAP) inference. We also consider the possibility that external data (e.g., from a ground truth distribution, like human generated data in LLMs) may also be supplied alongside self-generated data during repeated training. For this general setting, we derive the Fokker-Planck equations governing the stochastic

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dynamics of the model parameters during closed-loop learning. Finding the stationary state distribution of the Fokker-Planck equations, we find that the presence of priors/regularization and/or the injection of external data may prevent model collapse. In particular, we show that even adding one single data point from outside at each retraining step may be sufficient for avoiding model collapse in EFs. Surprisingly, these phenomena happen even in the limit where, at each retraining iteration, the size of the training set approaches infinity. This is surprising because, in this limit, the differences between MAP estimation and ML, as well as the effects of priors and regularization, become negligible in standard (nonclosed-loop) learning.

Closed-loop learning in EF—Consider the following dynamics in the space of parameters $\theta \in \mathbb{R}^d$. The dynamics starts at θ_0 which we take to be the result of fitting the model to some data. From the initial parameter vector, at iteration $t \geq 0$, (1) draw M independent samples, $\hat{s}^1(t), \dots, \hat{s}^M(t)$, from Eq. (1) with $\theta = \theta_t$ and m samples $\hat{s}^{M+1}(t), \dots, \hat{s}^{M+m}(t)$ generated from a distribution $q(s)$. Denoting these samples as $\hat{s}(t) \equiv (\hat{s}^1(t), \dots, \hat{s}^{M+m}(t))$, the likelihood is

$$p[\hat{s}(t)|\theta_t] = \prod_{i=1}^M f[\hat{s}^i(t)|\theta_t] \prod_{j=1}^m q[\hat{s}^j(t)].$$

(2) Given $\hat{s}(t)$ find the parameters

$$\theta_{t+1} \equiv \hat{\theta}(\hat{s}(t)) = \arg \max_{\theta} [p(\hat{s}(t)|\theta) p_0^u(\theta)]. \quad (2)$$

Set $t \rightarrow t + 1$ and repeat.

When $u = 0$ the parameters are inferred by ML, whereas for $u = 1$ inference relies on MAP estimation with a prior $p_0(\theta)$. This includes regularization schemes, e.g., Lasso [15], that can be represented by an appropriate prior. With $m = 0$ the dynamics is one in which data are generated uniquely from the model itself whereas with $m > 0$ new data generated from a distribution $q(s)$ are also included. The distribution q could also be the model generating the original data that determined θ_0 and reflect a *ground truth*. In this setting, at each step, t , the model performance can be quantified by the Kullback-Leibler (KL) divergence, $D_{\text{KL}}[q||f(\cdot|\theta_t)]$, whose continual increase can be used to determine model collapse.

The steps above yield a Markov dynamics on the parameters: at each step the new data is generated consistently with the data just learned. Below, we first study this dynamics for $m = u = 0$, that is ML and completely self-generated data. We then study the general case.

Closed-loop learning under ML ($m = u = 0$)—The components $\phi_a(s)$ in Eq. (1) are the sufficient statistics of the model. This means that for $m = u = 0$ given a sample $\hat{s}$, the ML estimates, $\hat{\theta}(\hat{s})$, depend solely on the empirical averages

$$\bar{\phi}_a(\hat{s}) = \frac{1}{M} \sum_{l=1}^M \phi_a(\hat{s}^l), \quad (3)$$

and are the solutions of the following set of equations:

$$\langle \bar{\phi}_a(s) \rangle_{\hat{\theta}} = \bar{\phi}_a(\hat{s}), \quad a = 1, \dots, d, \quad (4)$$

where $s = (s^1, \dots, s^M)$, for $s^l \in \mathcal{S}$ and $\langle \dots \rangle_{\hat{\theta}}$ is expectation with respect to $p(s|\theta) = \prod_{l=1}^M f(s^l|\theta)$.

In the closed-loop dynamical process described above, $\hat{s}(t)$ determines $\bar{\phi}_a(\hat{s}(t))$ and thus, via Eq. (4), determines $\theta_t = \hat{\theta}(\hat{s}(t))$ and the distribution $p(\cdot|\theta_t)$. Consequently, given $\hat{s}(t)$ and defining $\bar{\phi}_{a,t} \equiv \bar{\phi}_a[\hat{s}(t)]$, we can write,

$$\langle \bar{\phi}_{a,t+1} | \bar{\phi}_{a,t} \rangle_{\theta_t} = \bar{\phi}_{a,t}, \quad (5)$$

where $\langle \dots | \bar{\phi}_{a,t} \rangle_{\theta_t}$ is the average over all possible samples $\hat{s}(t+1)$ generated at fixed θ_t , which as described above is determined by $\bar{\phi}_{a,t}$.

By definition, Eq. (5) means that $\bar{\phi}_{a,t}$ is a martingale [16]. As shown in Feller [17] (see also Supplemental Material (SM), Sec. I [18] for details), a Markov chain $\bar{\phi}_{a,t}$ that is also a martingale in a bounded state space converges to absorbing states. In other words, the fluctuations of $\bar{\phi}_{a,t}$ vanish; i.e., $\phi_a(s)$ attains the same value for all data points $\hat{s}(t)$, and

$$\lim_{t \rightarrow \infty} f(s|\theta_t) = c \prod_a \delta(\phi_a(s) - \bar{\phi}_{a,\infty}) \quad (6)$$

with c a normalizing constant. While the ground truth can be any arbitrary distribution, even well fitted by f at $t = 0$, closed-loop learning prevents the model from representing anything but few possibilities. Since this distribution assigns non-zero probability only to a limited set of states, its KL divergence with a generic ground truth $q(s)$ is infinite and the model collapses. Below we explore this in more details for three example models.

Consider an Ising model defined over spin configurations $s = (\sigma_1, \dots, \sigma_N)$ with $\sigma_i = \pm 1$ as

$$f(s|\theta_1, \theta_2) = \frac{1}{Z} \exp \left[\theta_1 \sum_i \sigma_i + \frac{\theta_2}{N} \sum_{i < j} \sigma_i \sigma_j \right]. \quad (7)$$

With $\theta_2 = 0$ (no interaction), only $\phi_1(s) = \sum_i \sigma_i$ is relevant and $\bar{\phi}_1(\hat{s})$ converges to $\pm N$, with (see SM, Sec. I [18] for details)

$$\lim_{t \rightarrow \infty} P\{\bar{\phi}_{1,t} = N | \bar{\phi}_{1,0}\} = \frac{\bar{\phi}_{1,0} + N}{2N}. \quad (8)$$

Correspondingly, $\theta_{1,t}$ diverges to $\pm \infty$ and the distribution $f(s|\theta_1, \theta_2 = 0)$ converges to a fully polarized state where in

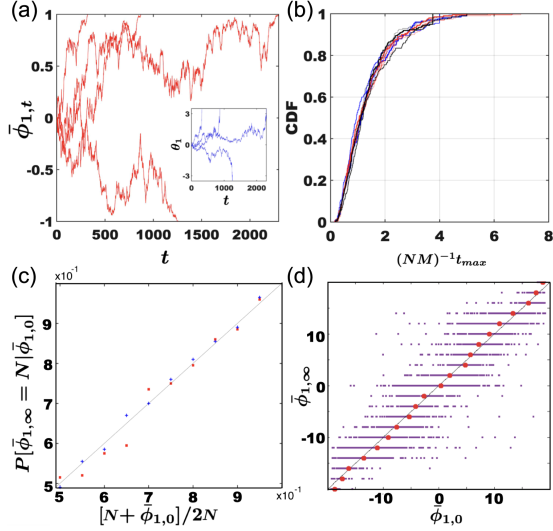


FIG. 1. Closed-loop dynamics and model collapse for $m = u = 0$ in the Ising model (Eq. (7)). (a)–(c) Refer to the case $\theta_2 = 0$ whereas in (d), $\theta_2 \neq 0$. (a) Example trajectories of $\phi_{1,t}$, starting at $\phi_{1,0} = 0$; $N = 20$ and $M = 50$. Corresponding trajectories for θ_1 are shown in the inset. (b) Cumulative distribution for the convergence time to fully polarized states normalized by NM for $N = 5, 10, 20$ (blue, red, black lines), and $M = 50, 100, 200$ (thinnest to thickest lines). (c) Probability of convergence to all spins up, as a function of $(1 + \bar{\phi}_{1,0})/2$ (see Eq. (8)); each point is calculated from 200 simulations with the same initial $\bar{\phi}_{1,0}$, with $N = 10$; $M = 50$ (red squares) and $M = 100$ (blue circles). (d) The final values of $\bar{\phi}_1$ vs initial values for $\theta_2 \neq 0$, for 100 different simulations with $N = 20$ and $M = 100$. θ_2 is set to zero at $t = 0$ but updated afterwards. Full circles are averages of $\bar{\phi}_{1,0}$ for the same value of $\bar{\phi}_{1,\infty}$.

the sample $\hat{s}$, either $\sigma_i^l = +1$ or $\sigma_i^l = -1$ for all l . Figure 1(a) shows examples of this dynamics. The empirical cumulative distributions of $t_{\max}$, the time it takes for the dynamics to converge, shown in Fig. 1(b), suggests that $t_{\max} \sim NM$. Figure 1(c) shows simulations results confirming Eq. (8).

If instead $\theta_2 \neq 0$, any $\bar{\phi}_{1,\infty} = N - 2k$ with $k = 0, 1, 2, \dots, N$, can be an absorbing state. For $|\bar{\phi}_{1,\infty}| < N$, this occurs because both θ_1 and θ_2 diverge, i.e. $\theta_2 \rightarrow -\infty$ with a finite ratio $\theta_1/\theta_2 = -\bar{\phi}_{1,\infty}/N$; see SM [18]. In this case, $f(s|\theta_1, \theta_2)$ converges to a uniform distribution on all configurations s with the same value of $\phi_1(s)$. Figure 1(d) shows that for $\theta_{2,t=0} = 0$, the limiting value of $\bar{\phi}_{1,\infty}$ is not far from the initial value $\bar{\phi}_{1,0}$.

To further illustrate model collapse in closed-loop learning, we next consider a spin model where θ_s is an independent parameter for each of the $d = 2^n$ spin configurations $s = (\sigma_1, \dots, \sigma_N)$. The corresponding operator in Eq. (1) is thus $\phi_s(s') = \delta_{s,s'}$. When θ_s are drawn from a Gaussian distribution, this model coincides with the *random energy model* [19]. Yet here we focus on the closed-loop learning setting with $m = u = 0$, where θ_{t+1} are the ML estimates inferred from M samples $\hat{s}(t)$ drawn i.i.d.

from the model with parameters θ_t . For $M \ll 2^N$ this is an over-parametrized model. The average of ϕ_s over the data $\hat{s}(t)$ is $\bar{\phi}_{s,t} = k_{s,t}/M$, where $k_{s,t}$ is the number of times configuration s occurs in the data $\hat{s}(t)$, and $f(s|\hat{\theta}(\hat{s}(t))) = (k_{s,t}/M)$. A state s which is not observed at time t , i.e., if $k_{s,t} = 0$, will thus never be observed forever after. The number of states with $k_{s,t} > 0$ is thus a non-increasing function of time. The asymptotic state will be one where $k_{s^*,t} = M$ for one state and $k_{s,t} = 0$ for all $s \neq s^*$, which correspond to the absorbing state with $\bar{\phi}_{s^*} = 1$ and $\bar{\phi}_s = 0$ for $s \neq s^*$. Furthermore, it can be shown that (see SM I [18])

$$\lim_{t \rightarrow \infty} P\{k_{s,t} = M | k_{s,0}\} = \frac{k_{s,0}}{M}. \quad (9)$$

Summarizing, closed-loop learning in EF based on ML and no fresh data ($m = 0$) leads to a reduction of the diversity of the data: in this case upon closed-loop learning, the model assigns zero probability to most states, even if they had nonzero frequency in the original dataset or under $q(s)$.

Continuous time dynamics in the general case— Consider now the general case with arbitrary m and $u \in \{0, 1\}$. Between two consecutive iterations, the difference $\theta_{t+1} - \theta_t$ is expected to be small, i.e., of order of the error on the ML estimator, $\delta\hat{\theta} \propto M^{-1/2}$. For large M , then, the evolution of θ_t is well approximated by a continuous time dynamics. Take the transition matrix

$$p(\theta_{t+1} = \theta' | \theta_t = \theta) = \sum_{\hat{s}} p(\hat{s} | \theta) \delta(\theta' - \theta_{t+1}(\hat{s})), \quad (10)$$

where $\theta_{t+1}(\hat{s})$ is given by Eq. (2) and the probability of $\hat{s}$ may depend both on $\theta = \theta_t$ and on q , as detailed in step 1 of the dynamics described above. For $M \rightarrow \infty$, the transition matrix is dominated by a sharp maximum close to the ML estimator $\hat{\theta}$. Expanding around this point (see SM, Sec. II [18] for details), to the leading order in M , θ_{t+1} is well approximated by a Gaussian variable with covariance $J^{-1}(\theta_t)/M$, where

$$J_{a,b}(\theta) = \frac{\partial^2}{\partial \theta_a \partial \theta_b} \log Z = \langle \phi_a \phi_b \rangle_{\theta} - \langle \phi_a \rangle_{\theta} \langle \phi_b \rangle_{\theta} \quad (11)$$

is the Fisher information matrix (FIM), and mean $\theta_t + A(\theta_t)/M$, where

$$A_a(\theta) = - \sum_b J_{a,b}^{-1} \frac{\partial U}{\partial \theta_b} \quad (12)$$

$$U(\theta) = \frac{1}{2} \log \det J - u \log p_0(\theta) + m D_{\text{KL}}[q || \theta], \quad (13)$$

where $D_{\text{KL}}[q || \theta] \equiv D_{\text{KL}}[q || f(\cdot | \theta)]$. Note that in the ML case ($m = u = 0$) the drift term (A_a) is nonzero. In the

rescaled time $\tau = t/M$, the vector of parameters satisfies the (Ito) stochastic differential equation [20] (see SM Sec. II [18] for details)

$$d\theta_\tau = A(\theta_\tau)d\tau + dW_\tau \quad (14)$$

$$\langle dW_\tau dW_{\tau'} \rangle = J^{-1}(\theta_\tau) \delta(\tau - \tau') d\tau. \quad (15)$$

This dynamics is fully consistent with the model collapse discussed in the previous section for $m = u = 0$. Indeed Eqs. (14) and (15) can be used to show that the evolution of the sufficient statistics $\varphi_a \equiv \langle \phi_a(s) \rangle_\theta$ is a purely diffusive dynamics (see SM, Sec. II [18] for details), eventually hitting and getting absorbed at the boundaries.

The continuous time dynamics for $m = u = 0$ indeed shows that closed-loop learning leads to a contraction of the configuration space. For example, for an Ising model with $\theta_2 = 0$, the dynamics of the magnetization $\mu = \langle \tilde{\phi}_1 \rangle_\theta / N = \langle \sum_i \sigma_i \rangle_\theta / N$ per spin is given by $d\mu = \sqrt{1 - \mu^2} dW$, where dW is the differential of a Wiener process. This dynamics clearly has two absorbing states at $\mu = \pm 1$. Indeed $\langle \mu^2 \rangle_{\theta_\tau} = 1 - (1 - \langle \mu^2 \rangle_{\theta_0}) e^{-\tau}$. More generally, one can consider the evolution of the entropy,

$$S(\varphi) = \log Z - \sum_a \theta_a \varphi_a, \quad (16)$$

as a function of the conjugate variables φ to show that the configuration space, indeed, shrinks. To see this, note that $-S(\varphi)$ is the Legendre transform of $\log Z$, in the conjugate coordinates φ , and hence $(\partial S / \partial \varphi_a) = -\theta_a$ and $(\partial^2 S / \partial \varphi_b \partial \varphi_c) = -J_{b,c}^{-1}$ [13]. The differential of the entropy is thus $dS = -\sum_a \theta_a d\varphi_a - D d\tau$. Consequently, the expected value of the entropy changes with time τ as

$$\langle S \rangle_{\theta_\tau} = \langle S \rangle_{\theta_0} - \tau D \quad (17)$$

and diverges to $-\infty$ as $\tau \rightarrow \infty$. This is a quantitative version of what we stated at the end of the last section: closed-loop learning for $m = u = 0$ reduces the diversity of the data. Importantly, Eq. (17) shows that the entropy reduces at a faster rate for higher dimensional models.

In the presence of priors and/or of samples drawn from $q(s)$ ($u = 1$ and/or $m > 0$), the asymptotic behavior of the dynamics can be studied by deriving the Fokker-Planck equation associated to Eq. (14) using standard techniques (see SM, Sec. II [18]). This shows that the dynamics converges to a stationary state

$$p_{st}(\theta) = c p_0^{2u}(\theta) e^{-2m D_{KL}[q||\theta]}, \quad (18)$$

provided the normalization constant c is finite. Consistent with the model collapse identified in the previous section, when $m = u = 0$ the right hand side in Eq. (18) is not normalizable. However, even the addition of one external

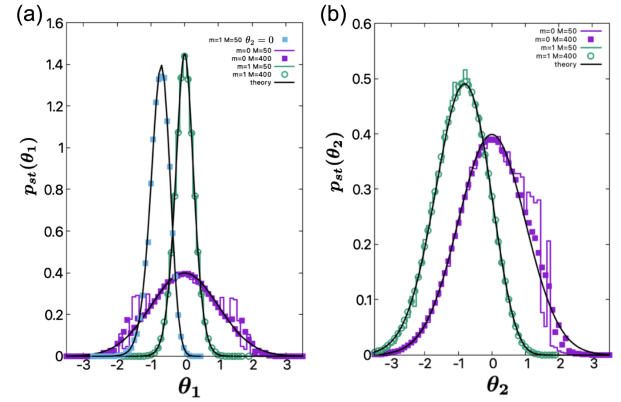


FIG. 2. Numerical simulations confirming Eq. (18) and prevention of model collapse for the Ising model. (a) The distribution of the field θ_1 . (b) The distribution of the coupling θ_2 . We consider three cases all with $N = 10$; (i) $\theta_2 = 0$ and $M = 50$, with one data point with $\sum_i \sigma_i = -6$ being added ($m = 1$) while $u = 0$ (ML); (ii) $\theta_2 \neq 0$, $m = 0$ and $M = 50$ and 400 , with Gaussian priors on both θ_1 and θ_2 with zero mean and variance 2 (which is equivalent to L_2 regularization with parameter $1/4$); (iii) $\theta_2 \neq 0$, $M = 50$ and 400 , with the same priors and the addition of one data point ($m = 1$) with $\sum_i \sigma_i = 0$. In all cases model collapse is avoided as predicted by Eq. (18) with a good agreement (see full black lines). Data is obtained from 10^7 iterations of the dynamics.

data point ($m = 1$) at each retraining step or turning on the prior ($u = 1$) may change this (see also SM, Sec. IIB [18]). For example, for the Ising model in Eq. (7) with $\theta_2 = 0$ under ML ($u = 0$) and with $m = 1$, model collapse will be avoided, as confirmed numerically in Fig. 2(a). In Fig. 2, we also numerically confirm the analytical expression in Eq. (18) for MAP ($u = 1$) for the Ising model when $\theta_2 \neq 0$, $m = 0$ and $m = 1$, showing that model collapse is again avoided.

In general, one can show that while in the absence of a prior ($u = 0$), adding external data (any $m > 0$) alone is sufficient for preventing model collapse, in the absence of such data ($m = 0$), using MAP ($u = 1$) alone is not (see SM, Sec. IIB [18]). An example where $u = 1$ alone ($m = 0$) does not prevent model collapse is a Poisson distribution with mean ϑ . This model with $\theta = \log \vartheta$ belongs to the EF. Assuming an exponential prior $p_0(\vartheta) = \lambda e^{-\lambda \vartheta}$ and with $q(s)$ a Poisson distribution with mean ϑ_0 , the stochastic differential equation that ϑ follows turns out to be that of the Cox-Ingersoll-Ross process [21]. With $m = 0$, the process does not converge to a stationary state, because Eq. (18) is not normalizable, and the system converges to the absorbing state $\vartheta_\infty = 0$ even with $\lambda > 0$. This can, however, be prevented even with a small amount of data from outside ($m > 0$) for which the dynamics converges to the stationary state of Eq. (18), which is a Gamma distribution with shape parameter $2m\vartheta_0$.

Discussion—We studied closed-loop learning for models in the exponential family, analytically deriving the equations governing the stochastic dynamics of the

parameters. We found that in this setting, as a result of the martingale property [Eq. (5)], model collapse occurs when inference is based on the maximization of the likelihood on data entirely generated by the model itself. We showed that the introduction of even one single data point generated from a ground truth distribution at each retraining step, or adopting MAP estimates or introducing regularization can avoid model collapse. In classical statistical theory, the difference between MAP ($u = 1$) and ML ($u = 0$) estimates vanishes for $M \rightarrow \infty$. Likewise, polluting a sample with a finite number m of data points drawn from a different distribution has no effect on the asymptotic behavior of statistical estimates. Our results, however, show that these conclusions do not hold in closed-loop learning.

Although our results were derived for the EF in the well-sampled regime, it is interesting to observe that they align, at a qualitative level, with observations reported in more complex cases [1–8]. In order to explore to what degree our results extend to more complex models, we studied closed-loop learning in restricted Boltzmann machines (RBMs; see SM, Sec. IV [18]), a building block of many real life applications [22–24]. In spite of the fact that the RBM is highly overparametrized and has a singular FIM that invalidates the expansion leading to Eq. (14), we find a qualitatively similar phenomenology, compatible with model collapse for $m = 0$, and its prevention with as little as $m = 0.01M$ external data added per closed-loop learning step. This is consistent with the fact that any smooth and bounded density can be approximated arbitrarily well by an element of a sufficiently high-dimensional exponential family [12,13].

Still, it remains to be seen how much our results extend to state-of-the-art large neural networks which are also overparametrized and operate in the undersampled regime. For example, heuristics used in training large neural networks may introduce some explicit or implicit regularization that likely influence the outcome of closed-loop learning. Furthermore, convergence of training in large neural network cannot be taken for granted and it may require very long times [25].

Given the above, our analysis is thus likely to provide a starting point for addressing issues such as implicit biases arising from algorithms, architectures, and/or initial conditions in over-parametrised artificial neural networks [26–28], providing intuition and testable prediction within an analytically controllable framework. Generalizing our analysis to models where FIM is not invertible using an approach such as that of [29], and studying the FIM in deep networks using techniques such as those of [30,31] may thus help understanding the precise dynamics of closed-loop learning in such networks.

Dynamics similar to closed-loop learning are present in many domains including word-of-mouth communication, repeated model based imitation [32–34] and rational herding [35], or cultural transmission across generations [36]. Preliterary cultural evolution has been based on one

generation learning from the previous one, leading to a corpus of “data” of, e.g., mythological nature that shares remarkable similarities across distant cultures [36]. In social contexts, individuals learn from the observed actions of peers, their resulting behavior is observed by others, and this has been argued to determine many collective phenomena [35]. In prenatal development, the brain develops to a large extent in the absence of external stimuli, driven by its own spontaneous activity [37,38]. What is common in these phenomena is that learning takes place using data generated by another learned model. Although the process discussed here is very simple and stylized, it offers a platform for studying these phenomena.

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Data availability—The codes used for numerical results of this Letter are available at [39].

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